

2-Mark Questions & Answers -- Data Structures Module 6: Sorting & Searching Techniques

Q1. Differentiate between Linear Search and Binary Search.

- * Linear Search: Unsorted/sorted arrays, sequential scan, $O(N)$ time.

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Q2. What is a Stable Sorting algorithm Give examples.

A sorting algorithm is Stable if it preserves the relative order of duplicate elements with equal keys.

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Q3. Define Hashing and Collision.

- * Hashing: Mapping a search key K to a table index $h(K)$ using a hash function.

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Q4. What is Linear Probing vs Quadratic Probing

- * Linear Probing: Probes linearly $h(k, i) = (h'(k) + i) \bmod m$. Suffers from primary clustering.

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